

AISHA OMAR FARAH

Backend & Distributed Systems Engineer

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SUMMARY

Results driven Backend & Systems Engineer with a B.Sc. in Computer Information Systems and specialized expertise in distributed systems, high throughput payment rails, and protocol infrastructure. Proven track record of architecting asynchronous REST/gRPC APIs, real time WebSocket engines, and automated CI/CD tooling using Node.js, Express, Python, Docker, and GitHub Actions. Experienced in containerized microservices, open source maintainer tooling, and smart contract architecture within cross functional engineering teams.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, TypeScript, Python, Node.js, Express.js, Django, FastAPI, React 19 | APIs & Protocols: RESTful APIs, gRPC, JSON RPC, WebSockets, JWT Authentication, Macaroon Authentication

Distributed Systems & Tooling: Layer 2 networks, payment channels, GitHub Actions, CI/CD workflows, Polar, Regtest, Internet Computer Protocol (ICP), smart contracts

Databases & Caching: PostgreSQL, MySQL, Redis, SQLite, Prisma ORM | DevOps & Tools: Docker, Docker Compose, Git, GitHub Actions, Postman, Vercel, Render

Frontend: React 19, TypeScript, JavaScript, Tailwind CSS, HTML5, CSS3

TECHNICAL PROJECTS

Layer 2 Payment Gateway — Full Stack Systems Implementation github.com/Aishagojo/e-commercepayment

- Engineered a non custodial checkout gateway enabling real time order settlement via cryptographic invoices, fully tested on a local testnet environment.
- Built an asynchronous Node.js/Express backend integrating node REST APIs via restricted invoice macaroons and TLS verification, with automated state polling and WebSockets.
- Developed a React 19/Vite storefront rendering QR invoices with countdown timers, clipboard utility hooks, and real time pending/paid UI state updates.
- Orchestrated a private two node Polar/Docker payment network (Merchant and Customer nodes) and verified settlement using an automated test runner.

CoreGate: Bitcoin Core Triage & CI/CD Engine — Backend & Tooling Engineer github.com/Zeegaths/core-gate

- Architected a deterministic Node.js rules engine and GitHub Actions workflow to score incoming Bitcoin Core pull requests against official contribution guidelines in under 30 seconds.
- Engineered a 3 lane triage system and risk profiling algorithm for security critical consensus code (src/consensus/, src/crypto/), incorporating AI slop detection and automated feedback.
- Won 1st Place, Open Source Track, at the Bitcoin++ Nairobi Hackathon (2026), sponsored by Btrust.

PesaPipe: Cross Border Remittance Gateway — System Architecture (Ongoing) github.com/Aishagojo/Pesapipe

- Designing a headless remittance payment gateway bridging Layer 2 blockchain settlement with Kenyan mobile money ecosystems (KES/USSD).
- Building lightweight terminal based transaction drivers and REST endpoints for low bandwidth mobile execution.

Decentralized Visitor Management System — Web3 Systems Developer github.com/Team-Z-ICP/visitor_system

- Engineered smart contract canisters on the Internet Computer Protocol (ICP) for decentralized access control and immutable check in logs.
- Won 1st Place at the ICP Hackathon 2024 for canister security architecture and identity verification.

WORK EXPERIENCE

Technical Operations Intern, Comex Saku Business Park Nairobi, Kenya | Jun 2025 – Aug 2025

- Deployed and provisioned 15+ network workstations, improving operational efficiency across office workflows by 25%.
- Diagnosed and resolved 50+ software, hardware, and network issues weekly, maintaining 99%+ system availability for staff.
- Monitored infrastructure performance and resolved technical faults, reducing system downtime by 30%.

IT Systems Intern, Ministry of Industry, Trade and Investment Nairobi, Kenya | Oct 2024 – Nov 2024

- Restructured internal helpdesk workflows, cutting technical issue response and resolution times by 40%.
- Audited and maintained a centralized digital inventory tracking 200+ hardware and software assets.
- Collaborated with the core ICT team to sustain 99% network uptime across critical government operations.

EDUCATION

B.Sc. in Computer Information Systems, Kenya Methodist University Nairobi, Kenya | 2025

GPA: 3.08 / 4.00 | Core Coursework: Advanced Programming, Data Structures & Algorithms, Database Systems, Software

Engineering, Network Security

CERTIFICATIONS & RESIDENCIES

- Bitcoin, AI & Open Source Residency — Dada Devs, Dada Hub (2026)
- Backend Development Specialization — ALX Africa (2025)
- AI Career Essentials — ALX Africa (2025)
- Professional Foundation Programme — ALX Africa (2024)